

Shaurya Singh Rawat

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Education

Cranfield University – MSc in Aerospace Computational Engineering

September 2022

Delhi Technological University – B.Tech in Mechanical Engineering

June 2021

Experience

Co-founder & Engineering Lead, Maviyom Aviation – Delhi, India

March 2024 - Present

- Led systems integration, field testing, and operational deployment activities for fixed-wing VTOL, tethered UAS, and autonomous aerial systems during defence and government platform evaluations.
- Developed and validated a hybrid fixed-wing VTOL ISR platform during high-altitude operations in Ladakh, conducting flight trials from ~4955 m ASL launch altitude and cruise profiles up to 1000 m AGL to evaluate endurance, failsafe behaviour, and EO target identification performance.
- Integrated and ruggedized airborne communications relay payloads onto tethered UAS platforms for sustained 24-hour field operations in 48°C ambient and >60°C ground-temperature conditions.
- Developed and demonstrated a fiber-optic-guided loitering munition concept integrating operator in the loop, real-time guidance, and contested-environment operation during multi-vendor field evaluations.
- Developed and deployed FPV trainer UAS platforms and operational training labs for Indian MoD end users, integrating airframes, avionics, communications, and support systems for pilot training applications.
- Designed custom avionics support hardware and proof-of-concept visual odometry pipelines using KiCAD, onboard computer vision, and autonomous navigation methods.

Projects

Hypersonic Inlet Buzz at Mach 5

MSc Dissertation

- Conducted 2D URANS simulations of hypersonic inlet buzz at Mach 5 in ANSYS Fluent using a $k-\omega$ SST model and third-order MUSCL discretisation on a 296k element structured grid.
- Simulated open and blocked isolator configurations (blocking ratios 0.136 and 0.34); identified big buzz oscillation frequencies of 1,548 Hz and 1,444 Hz respectively, both consistent with the quarter-wave resonator mode of the intake-isolator system (within 6.5% of theory).
- Validated qualitatively against experimental schlieren data, confirming URANS captured low-frequency “big buzz” behaviour but not higher-frequency “little buzz” oscillations.

Fluid-Structure Interaction of Civil Aircraft During Rapid Decompression

MSc Group Project

- Developed a two-fidelity FSI methodology to simulate explosive decompression of an Airbus A320 fuselage, coupling CFD and FEA solvers via ANSYS System Coupling.
- Modelled fuselage structural response and cabin acoustics across multiple panel layouts and opening sizes, identifying 15 Hz buffeting with SPLs of 131-149 dB depending on breach size.
- Conducted grid convergence study (GCI) to validate the simulation framework; confirmed structural stresses and deflections remained within component limits.

Aerodynamics Lead - Team Defianz Racing

Formula Student

- Led the aerodynamics division of a ~30-member Formula Student team, developing a full aero package — wings, diffuser, and composite bodywork, from CFD concept to race-ready hardware.
- Designed aerodynamic surfaces in **STAR-CCM+** and **SolidWorks**, achieving a **10% increase in downforce coefficient** with only a **4% vehicle mass penalty**.
- Transitioned composite manufacturing from wet layup to resin infusion CFRP, producing smoother hollow aerodynamic structures with improved structural consistency and mass efficiency.
- Conducted thermal modelling and FMEA studies for the team’s electric vehicle cooling system concept, receiving the **Best FMEA Award** at the Formula Bharat Electric Vehicle Challenge.

Skills

CAD & Simulation: SolidWorks, ANSYS Fluent, ANSYS Mechanical, Star-CCM+, ICEM CFD, Abaqus, KiCAD

Programming: Python, MATLAB, C++